CLOUD COMPUTING

Cloud computing is the delivery of computing services (servers, storage, databases, networking, software, analytics, AI) **over the Internet** ("the cloud") on a **pay-as-you-go** basis. Instead of buying and maintaining your own hardware, you rent what you need from a provider such as **AWS, Microsoft Azure or Google Cloud**.

> **Example:** Own generator vs. electricity from the national grid: you do not build a power plant, you plug in and pay for the units you use. Cloud is IT "utility" service.

Evolution of Cloud Computing

Era	Description
1950s-60s	**Mainframes** and time-sharing: many users share one big computer through terminals
1970s-80s	**Personal computers** and early networks; virtual machines (IBM VM)
1990s	**Client-server** and the Internet; web hosting; application service providers (ASP); Salesforce (1999) delivers software over the web
Early 2000s	**Virtualization** matures; **grid and utility computing** ideas; data centers become efficient
2006	**Amazon Web Services** launches S3 and EC2, the start of modern public cloud
2008-2010	**Google App Engine, Microsoft Azure** (2010), OpenStack; cloud becomes mainstream
2010s	SaaS boom (Microsoft 365, Google Workspace), DevOps, containers, mobile apps
2020s	Hybrid and multi-cloud, serverless, edge computing, AI and machine learning in the cloud

Before Cloud vs With Cloud

Point	Before cloud (traditional on-premises)	With cloud
Hardware	Buy and own servers, storage, network devices	Rent from provider; no hardware purchase
Cost model	Big upfront capital expense (CAPEX)	Pay-as-you-go operational expense (OPEX)
Setup time	Weeks or months to order and install	Minutes to deploy a server

Point	Before cloud (traditional on-premises)	With cloud
Scalability	Difficult; buy for peak demand (waste when idle)	Scale up or down on demand automatically
Maintenance	Own team handles power, cooling, hardware repairs, patches	Provider handles the physical infrastructure
Location	Limited to your own data center	Global regions; access from anywhere
Disaster recovery	Expensive second site	Built-in replication and multi-region options
Updates	Manual, slow upgrades	Continuous, managed updates
Capacity planning	Must predict future needs	Elastic resources

Key Characteristics of Cloud

The US standard body **NIST** defines five essential characteristics:

Characteristic	Explanation
On-demand self-service	Users create resources themselves through a portal or API without calling the provider
Broad network access	Services are available over the network from any device (laptop, phone)
Resource pooling (multi-tenancy)	The provider shares resources among many customers with logical separation
Rapid elasticity	Resources are quickly increased or decreased as needed; appear unlimited
Measured service	Usage is monitored and billed like electricity (per hour, per GB, per request)

Cloud Deployment Models

Model	Description	Examples / use
Public cloud	Services offered over the public Internet by a provider; resources shared among many customers	AWS, Azure, Google Cloud; startups, web apps, development / testing

Model	Description	Examples / use
Private cloud	Cloud infrastructure used by a single organization (on-premises or hosted); more control and security	Banks, governments; VMware / OpenStack / Azure Stack
Hybrid cloud	Combination of private (or on-premises) and public cloud with data and apps moving between them	Keep sensitive data on-premises, use public cloud for peak load / DR
Community cloud	Shared by several organizations with common concerns (security, compliance)	Government agencies, universities, healthcare consortia
Multi-cloud	Using services from more than one public cloud provider	Avoid lock-in, use the best features from each

Cloud Service Models

Model	Provider manages	You manage	Examples
IaaS (Infrastructure as a Service)	Physical servers, storage, networking, virtualization	OS, middleware, runtime, applications, data	Amazon EC2, Azure Virtual Machines, Google Compute Engine
PaaS (Platform as a Service)	Infrastructure + OS + runtime + middleware	Applications and data	Azure App Service, Google App Engine, AWS Elastic Beanstalk
SaaS (Software as a Service)	Everything: infrastructure, platform and the application	Only your data and user settings	Microsoft 365, Gmail, Salesforce, Zoom
FaaS / Serverless	Everything including servers and scaling	Only your code (functions)	AWS Lambda, Azure Functions

> **Example:** Pizza as a service. **On-premises** = make pizza at home. **IaaS** = you buy the oven and kitchen space, and you cook. **PaaS** = ingredients and oven ready, you add toppings. **SaaS** = you just eat the pizza delivered to you.

Shared responsibility model

Security in the cloud is **shared**. The provider is responsible for security **of** the cloud (data centers, hardware, core services). The customer is responsible for security **in** the cloud (data, identities, access control, configuration, applications). The more you move from IaaS toward SaaS, the more the provider handles, but **you are always responsible for your data and for who can access it**.

Advantages of Cloud

- **Cost efficiency:** no big upfront hardware cost; pay only for what you use.
- **Scalability and elasticity:** handle sudden traffic easily.
- **Speed and agility:** deploy servers and applications in minutes.
- **Reliability and availability:** multiple data centers and regions, built-in redundancy.
- **Disaster recovery and backup:** easier and cheaper than building a second data center.
- **Global reach:** deploy close to users anywhere in the world.
- **Automatic updates and maintenance** handled by the provider.
- **Access from anywhere** supports remote work and collaboration.
- **Innovation:** quick access to AI, analytics, IoT and big data services.
- **Security tools and compliance certifications** from large providers.

Challenges of Cloud Computing

- **Security and privacy:** data is outside your direct control; misconfigured storage and weak identity management cause breaches.
- **Compliance and data sovereignty:** laws may require data to stay in a country (important for banks and governments).
- **Downtime and outages:** you depend on the provider and the Internet connection.
- **Vendor lock-in:** moving away from a provider can be hard and costly.
- **Cost management:** forgotten resources, data transfer (egress) charges and poor planning can cause surprise bills (FinOps is needed).
- **Limited control and customization** compared with your own infrastructure.
- **Latency and bandwidth:** performance depends on distance and the network.
- **Skills gap:** staff need cloud architecture and security skills.
- **Integration and migration complexity** with legacy systems.
- **Shared responsibility misunderstandings.**

Zoom: Cloud Adoption Strategy

Zoom is a good example of how a company uses cloud as a strategy. Zoom's video platform must handle very large and sudden changes in demand, and it needs high performance and reliability worldwide.

Key points of the strategy

- **Hybrid / multi-cloud approach:** Zoom combines its **own co-located data centers** (to control the latency-sensitive video traffic) with **public cloud providers** such as **AWS** and **Oracle Cloud Infrastructure (OCI)** for extra capacity and global reach.
- **Elastic scaling for sudden demand:** when remote work and online learning exploded in early 2020, daily meeting participants grew from around 10 million to over 300 million within a few months. Cloud capacity made it possible to add resources fast.

- **Global distribution:** many data centers and cloud regions near users to reduce latency and improve video quality.
- **Avoiding single-provider dependence:** using multiple providers reduces vendor lock-in and increases resilience and bargaining power.
- **Security and privacy improvements:** after early security criticism, Zoom added end-to-end encryption options, stronger default settings and regional data controls.
- **SaaS delivery model:** customers simply use Zoom through an app or browser; Zoom manages the infrastructure behind it.

Lessons from the case

- Cloud helps when **demand is unpredictable**.
- A **hybrid** model lets you keep control where performance is critical and use the cloud where flexibility is needed.
- Cloud adoption needs strong **security governance** from the beginning.